

KIA ORA! WELCOME TO THE PATIENT EMPOWERMENT PROGRAM FOR PROSTATE CANCER PATIENTS

This program aims to improve the mental health, physical health, and quality of life in men with a prostate cancer diagnosis while reducing some of the side effects from treatments. If you have not yet started treatment, these components are designed to get you in healthier shape, both physically and mentally, to optimize your post-treatment recovery. The program, based on evidence from the latest science and health research, will motivate you to create new healthy habits centered around the following five components:

- 1) Physical exercise – aerobic and strength training
- 2) Dietary and nutrition recommendations
- 3) Relaxation and breathing exercises for stress reduction and emotional regulation
- 4) Pelvic floor exercises (Kegels) to decrease urinary problems
- 5) Social support/connection advice and opportunities

Daily emails over the next 6-months will slowly deliver education, recommendations, and resources on the above components. We encourage you to engage with all components of the program, however there will likely be some components that benefit you more than others, so please take from the program what helps you the most.

Please review the documents in this package before you start the program.

WHAT TO EXPECT AS YOU BEGIN:

On your start date, you will receive your “PC-PEP Day 1” email. A staff member will inform you of your start date. You will receive an email every day for the next 182 days. In those emails you will find a link to a YouTube video with a daily greeting from Dr. Ilie and Dr. Rutledge providing specific education and outlining what to do that day. **Please try to view these videos every day.**

You will receive an email with a link to a short survey every Sunday (starting one week from your start date). This will ask you about your participation in each component over the past week (e.g. how many minutes of strength exercise did you do). Please try to keep track of how much you engage in the various components.

Find additional resources at: <https://pcpep.org/pc-pep-program-new-zealand/>. The **confidential** password is **1414**.

To speak with one of our Volunteer Mentors (previous participants and prostate cancer survivors), visit the website above to view their profiles and contact information.

Please do not hesitate to contact us at any time with questions or concerns:

New Zealand Coordinator

Email: PCPEP@PROSTATE.ORG.NZ

Phone: 0800 477 678



Email Troubleshooting

As part of this program, you will receive an email every morning for 6 months. In the event you do not receive a daily email, or begin to experience issues with receiving any of your emails, **please notify us immediately** at PEP@nshealth.ca.

The success of the program relies on you receiving these emails daily.

This rare occurrence is most commonly the result of the email provider flagging our emails as SPAM. It is most commonly an issue with institutional or government email addresses who have more strict firewalls to limit emails they categorize as spam.

To prevent this from happening, please add PEP@nshealth.ca to your email account contacts list. We strongly advise you to do this now to minimize the chance of this issue happening to you in the future.

Regards,

The PC-PEP team



Exercise Safety Guidelines

Physical activity is a safe and effective way for people of all ages to improve their health. Research has shown that physical activity can improve side effects after cancer treatments, quality of life, and help reduce anxiety, depression, and fatigue. It also may help people live longer by reducing the chance of cancer recurrence.

As part of this program, we want you to work up to completing 150 minutes of physical activity each week. We also want you to reduce sedentary behaviour, such as sitting for long periods of time.

Each week, we will ask you to aim for four+ days of aerobic exercises of your choice, like walking or swimming, and two days of strength resistance exercising. The strength resistance workouts have been designed by a Certified Exercise Physiologist specifically for people with cancer.

Exercising does come with some risks that you need to be aware of. The most common risks include muscle, bone, or joint injuries. The risk of cardiac events, like a heart attack, also increases slightly during strenuous physical activity, especially in people with heart conditions. However, when you exercise safely, the potential benefits of exercise outweigh the risks. This guide will give you some tips to ensure you are engaging in the workouts safely and effectively.

Be mindful of any of the following symptoms before, during, and after your workouts:

- Chest pain, pressure, or discomfort
- Rapid and/or irregular heartbeat
- Difficulty breathing
- Chills, headaches, or dizziness
- Blurred vision
- Intense fatigue and muscle soreness
- Bone and/or joint pain

If you experience any of these symptoms, you should talk to your doctor before continuing with the workout plan.

Listen to your body

If your body does not feel ready to engage in physical activity, do not force yourself. Muscle soreness and fatigue is normal after exercising, but if you have any of the symptoms listed above then do not proceed with the workout. It may take time to get used to the workouts if you have not exercised regularly for some time. Do what you can and slowly build up your capacity.

Follow the instructional videos

Over the six-month period of the program, we send guided videos on how to perform the strength resistance exercises of our workout plan. To make sure you are performing the exercises safely, watch the videos explaining the purpose and proper form of each exercise. If done properly, the exercise may feel difficult but should not be painful.

Stay Hydrated

It is important that you always stay hydrated, especially when you are exercising. This will help you feel energized and prevent injury. Drink plenty of fluids before, during, and after exercise. At least 500ml (2 cups) of water is generally recommended before exercise, and 150ml every 15 minutes during exercise. The more you sweat, the more water you need drink.

Rest if You are Ill, Injured, or Recovering from Surgery

Take time to rest if you are feeling sick or get injured during the six-month program. Your body needs all its energy to heal if it is sick or hurt, so give it the necessary time to return to normal. Once you are feeling consistently better, you can continue your workouts.

If you get injured while participating in any of the activities of this program, please let us know right away.

If you have surgery during the program, stop exercising until your surgeon says that you can resume. This may take 6 weeks or more after your surgery. When you do resume exercising, start slowly.

Make Sure You Warm-Up

Doing a proper warm-up before exercising is a terrific way to get the most out of your workout and prevent injury. The purpose of the warm-up is to raise our body temperature and prepare our muscles for the activity ahead. Warm-ups should include low intensity activities such as brisk walking and stretching. This can be followed by some easy practice repetitions of the exercise you are about to do to activate those specific muscles.

Wear Appropriate Clothing and Use the Right Equipment

Make sure you are wearing clothes that do not restrict your movement too much when exercising. Stretchy items work best to allow full range of motion when being physically active. If you are participating in an athletic activity outside of our workout plan, be sure to wear the necessary equipment. These can include running shoes, shin pads, helmets, etc.

Always inspect your resistance bands for tears before using them. Feel free to swap out the resistance bands that we provide with your own equipment, such as light dumbbells. If you do use your own equipment, make sure you use them safely. Consider getting guidance from an exercise professional.

We hope this guideline helps you to exercise safely and enjoy your workouts. Remember that fitness is a process, so don't push yourself too hard, too fast. Enjoy!!

Kōrero Korikori 1

Activity Statement 1



Sit less, move more!
Break up long periods of sitting

Reasons for the recommendation

Based on the NHMRC quality rating system, the evidence for recommending sitting less and moving more is convincing.

The evidence indicates that sitting for long periods during waking time increases the risk of poor general health, type 2 diabetes and a range of weight-related health conditions such as obesity. Prolonged sitting increases insulin resistance and circulating blood sugar levels, and may increase the risk of type 2 diabetes independently of physical activity.

- People who sit for less than eight hours a day in total are more likely to have better health outcomes than those who sit for more than eight hours a day.

Sitting for less time can improve a person's health, even if they are already physically active.

- Sitting for long periods increases the risk of developing non-communicable diseases. Its effect is independent of the time a person spends being physically active.
- Doing 2½ hours of physical activity each week may not be enough to offset the negative effects of prolonged sitting time.

There is not enough evidence to identify the effect of too much sitting on cancer, cardiovascular disease and depression.

Source: Brown et al (2012)

Kōrero Korikori 2

Activity Statement 2



Do at least 2½ hours of moderate or 1¼ hours of vigorous physical activity spread throughout the week



Pregnant women should aim to do 2½ hours of moderate-intensity physical activity spread over at least 3 days per week (preferably some activity every day)

Reasons for the recommendation

Based on the NHMRC quality rating system, the evidence is:

- convincing for the recommendation to do at least 2½ hours of moderate or 1¼ hours of vigorous physical activity each week
- probable for the recommendation to spread physical activity throughout the week
Doing regular physical activity³⁹ helps to reduce the risk of developing: type 2 diabetes (whether or not you lose weight); colon, endometrial and postmenopausal breast cancer; and anxiety, depression and stress. It can also help people diagnosed with these conditions to manage them.

Regular physical activity also reduces the risk of dying early from any cause.

Spreading physical activity across the week has the greatest health benefits.

Growing evidence indicates that people can get extra health benefits (for example, a reduced risk of heart disease) by doing some vigorous physical activity rather than just moderate physical activity.

Source: Brown et al (2012)

³⁹ Regular physical activity is doing at least 2½ hours of moderate physical activity or 1¼ hours of vigorous physical activity or an equivalent mix of moderate and vigorous physical activity spread throughout the week.

Kōrero Korikori 3

Activity Statement 3



For extra health benefits, aim for 5 hours of moderate or 2½ hours of vigorous physical activity spread throughout the week



Pregnant women should seek advice from a health care professional with specialist knowledge about the impact of vigorous-intensity activity if competing in events or if exercising significantly more than Activity Statement 2

Reasons for the recommendation

Based on the NHMRC quality rating system, the evidence is:

- convincing for the recommendation to do at least 5 hours of moderate or 2½ hours of vigorous physical activity each week for extra health benefits
- probable for the recommendation to spread physical activity throughout the week (Brown et al 2012).

New Zealand adults can achieve extra health benefits by doing 5 hours of moderate or 2½ hours of vigorous physical activity, or an equivalent mix of both, each week. For example, by doing this higher amount of activity, they can do even better at:

- reducing the risk of developing type 2 diabetes
 - reducing the risk of gaining excess weight
 - reducing depressive symptoms and anxiety
 - increasing quality of life and wellbeing.
-

Kōrero Korikori 4

Activity Statement 4



Do muscle strengthening activities on at least two days each week



Pregnant women may also benefit from doing stretching and pelvic floor muscle training daily

Reasons for the recommendation

Based on the NHMRC quality rating system, the evidence is either convincing or probable for the recommendation to do some muscle strengthening activities on at least two days each week.

Regular muscle strengthening and weight-bearing activities help to reduce the risk of developing metabolic syndrome, pre-diabetes, osteoporosis and osteoarthritis and of having falls and fractures.

Muscle strengthening activities can be useful in managing osteoporosis and osteoarthritis.

Source: Brown et al (2012)

Some evidence, albeit limited as assessed using the GRADE system, is that pregnant women should do pelvic floor muscle training⁴⁶ daily to reduce urinary incontinence.

⁴⁶ For the greatest benefit, women should get instruction on the proper technique from a pelvic health physiotherapist.

Workout A

Shoulder Press		Stand or sit on the edge of a chair. Position your feet on the middle of a band. Holding onto the handles, turn your arms so your palms are facing away from you and then extend the arms and press straight overhead. Reverse the movement by lowering the arms back down in a controlled motion to the starting position. Stand/sit up tall and engage your core throughout the entire movement.
Air Squat		Stand with feet shoulder width apart. Bend the knees and hinge at the hips like you are sitting back into a chair until your knees reach 90 degrees. Return to the starting position and repeat. Ensure your weight is in yours and your knees are stacked directly over your ankles. Do not let the knees cave in and make sure your chest is being kept upright. Stay within a pain free range of motion.
Hydrant		Start in a table top position; wrists directly under shoulders, knees under hips. Open one leg out to the side (like a dog at a fire hydrant) feeling a squeeze in your glute. Return to the table top position and complete all reps before switching to the other side. The height of the leg lift is not important, your focus should be on maintaining proper form.
Chest Press		Anchor a band behind you at chest height. Stand far enough away that there is some tension on the band. Hold the band in either hand so your palms face down and extend your arms out in front. Bend your elbows to return to the starting position and repeat.
Calf Raise		Stand with feet hip width apart on a flat surface or at the edge of a step. Lift heels to rise up onto the balls of your feet. Lower back down and repeat. To advance this exercise, try holding weights in your hands by your side or try it using one foot at a time.
Front Plank		In a prone position (lying face down), position your elbows directly under your shoulders. Using either your knees or your toes as a pivot point. Ensure your body is in one straight line from the shoulders to the pivot point. Rise up into a plank and hold this position for the prescribed length of time.
Triceps Extension		Anchor a band in front of you at chest height. Stand far enough away that there is some tension on the band. Hold the band in either hand so your palms face down and extend your arms behind you to create a contraction in muscles on the back of your upper arm. Keep your elbows tucked into your rib cage. Return to start and repeat.
Side Lunges		Stand on a band, cross the handles over holding one in each hand. Slightly bend your knees and lower into a squat. Keep your feet slightly internally rotated while side stepping against the band's resistance. Side step 8-12 times in one direction, and then side step back with the other foot leading.
Cat Cow		Start in a table top position; wrists directly under shoulders, knees under hips. On the inhale, swoop your low back so that your belly is pushed toward the floor, while lifting your chest and bring your gaze to the horizon (similar to a cow in the pasture). Hold for 1-2 breathes. On the exhale, reverse the pose so that your low and upper back is arched like a bridge. Draw your belly button to your spine, while tucking your chin to your chest (similar to a cat stretching in the sun).
Level 1	15 seconds 'work' : 30 seconds 'rest'; Each round consists of nine exercises. Perform each exercise twice before moving on to the next. Once you have completed all nine exercises, start again. If you prefer to count repetitions, 15 seconds ≈ 8-12 repetitions. The complete workout should take approximately 27 minutes to complete.	
Level 2	20 seconds 'work' : 20 seconds 'rest'; Each round consists of nine exercises. Perform each exercise twice before moving on to the next. Once you have completed all nine exercises, start again. If you prefer to count repetitions, 20 seconds ≈ 12-15 repetitions. The complete workout should take approximately 24 minutes to complete.	
Level 3	30 seconds 'work' : 15 seconds 'rest'; Each round consists of nine exercises. Perform each exercise twice before moving on to the next. Once you have completed all nine exercises, start again. If you prefer to count repetitions, 30 seconds ≈ 16-24 repetitions. The complete workout should take approximately 27 minutes to complete.	
Level 4	45 seconds 'work' : 12 seconds 'rest'; Each round consists of nine exercises. Perform each exercise twice before moving on to the next. Once you have completed all nine exercises, start again. If you prefer to count repetitions, 45 seconds ≈ 24-36 repetitions. The complete workout should take approximately 35 minutes to complete.	

Workout B

Row		Anchor a band in front of you at chest height. Sit or stand far enough away that there is some tension on the band. Hold the band in either hand so your thumbs are pointing to the ceiling. Pull the band towards your body, elbows stay high and wide, feeling a squeeze in the muscles between your shoulder blades (upper back). Return to start and repeat.
Wall Sit		Stand with your back against a wall. Lower into a wall-supported squat. Your feet should be positioned far enough from the wall to allow for a 90 degree bend in your knees. Make sure your hips, knees and ankles maintain proper alignment.
Bird Dog		Start in a table top position; wrists directly under shoulders, knees under hips. Slowly, with control, extend opposite arm out in front, and opposite leg behind. Hold for slight pause and return to starting position. Repeat on opposite side. Throughout this exercise, maintain a neutral spine, with your head and neck aligned. Gaze on the floor.
Biceps Curl		Stand with your feet shoulder width apart on a band. Hold each end of the band in your hands with palms facing up. Keeping your elbows by your side, raise the handles toward your shoulders. Slowly lower down to starting position and repeat.
6-point Lunge		Using you left leg, step forward into a lunge. Return to standing. Reverse step into a lunge. Return to standing. Step to the side. Return to standing. Repeat this routine using your right leg.
Reverse Crunch		Lying flat on your back with your knees bent and feet flat on the ground. On an inhale, slowly pull your knees towards your chest, using your upper abdominals. On the exhale, slowly lower your knees until you feel your lower back arching away from the floor (end range of motion). Inhale, repeat.
Lat Pull		Hold a band overhead, slightly in front of you at a 45 degree angle with your hands approximately shoulder width apart. Lower the band down to the armpit level, while pulling your hands apart, so they make a rainbow shape. Reverse the movement, by raising your arms back up, while the hands return towards each other. Repeat.
Hip Lifts		Lying flat on your back with your knees bent and feet flat on the ground. On an inhale, press through your heels and raise your hips towards the ceiling, by driving your weight into your heels. Hold for a slight pause at the top and return to your starting position. Repeat.
Windshield Wipers		Lie flat on your back with your arms in a "T" position. Keeping your feet on the floor and knees together, with control, twist your hips so that your knees fall towards the floor. Using your lower abdominals and oblique muscles, return your knees to midline and repeat on the other side. Your head and shoulders should remain in contact with the floor. To make this exercise more challenging, try lifting your feet off the floor while maintaining a 90 degree bend in your knees.
Level 1	15 seconds 'work' : 30 seconds 'rest'; Each round consists of nine exercises. Perform each exercise twice before moving on to the next. Once you have completed all nine exercises, start again. If you prefer to count repetitions, 15 seconds ≈ 8-12 repetitions. The complete workout should take approximately 27 minutes to complete.	
Level 2	20 seconds 'work' : 20 seconds 'rest'; Each round consists of nine exercises. Perform each exercise twice before moving on to the next. Once you have completed all nine exercises, start again. If you prefer to count repetitions, 20 seconds ≈ 12-15 repetitions. The complete workout should take approximately 24 minutes to complete.	
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Ngā panoni tikanga kai, tikanga korikori tinana ka tūtohu mā ngā pakeke o Aotearoa

Recommended dietary and physical activity changes for New Zealand adults

Based on the evidence we considered and the resulting Eating and Activity Statements in these Guidelines, we recommend some changes² to the eating and activity practices of New Zealand adults. Tables 2 and 3 below summarise these changes.

Table 2: Recommended dietary changes for New Zealand adults

↓ Limit	↑ Increase
Processed meat	Vegetables and fruit
Red meat*	Legumes
Drinks and foods with added sugar	Fish and other seafood
Highly processed foods that are high in refined grains, saturated fat, sugar and salt	Nuts and seeds
	Whole and less processed foods
* Eat less than 500 g cooked red meat per week (equivalent to 700–750 g when raw).	
↻ Exchange	For
Refined grains	Whole grains and high fibre
Butter	Unsaturated vegetable oils and oil-based spreads
Higher-fat milk products	Low-fat milk products
Sugar sweetened beverages	Water, low-fat milk, diet drinks

² These recommended changes are based on data on average and common levels of food consumption and physical activity. Some New Zealanders may be already meeting some or all of the recommended intake and activity levels.

Table 3: Recommended activity changes for New Zealand adults

↓ Limit	↑ Increase
Prolonged sitting Prolonged screen time	Snackactivity** Light-intensity activity** Moderate-intensity activity** Vigorous-intensity activity** [∞] Muscle strengthening activity**
** See Glossary for definitions.	
↻ Exchange	For
Driving short distances	Walking, cycling or scooting [∞]
Prolonged driving	Regular breaks during driving
Taking a lift or escalator	Using the stairs
Prolonged sitting or screen time	Light activity such as standing and walking
Light activity	Moderate activity
Moderate activity	Longer or more frequent moderate activity Higher-intensity sessions [∞]

[∞] If pregnant, discuss these activities with your midwife or doctor.

Note: The concept for describing recommended changes in Tables 2 and 3 above, and Tables 4 and 5 in later sections, comes from the *Nordic Nutrition Recommendations 2012: Integrating nutrition and physical activity* (Nordic Council of Ministers 2014).

Eating Statement 1

Enjoy a variety of nutritious foods every day including: plenty of vegetables and fruit



Reasons for the recommendation

Vegetables and fruit provide vitamins, minerals and dietary fibre as well as many other phytonutrients (beneficial chemicals found in plants).

- Examples include folate in green leafy vegetables; pro-vitamin A (carotenoids) in yellow, orange, red and green vegetables; and potassium in a wide range of vegetables and fruit.
- Eating vegetables and fruit:
 - can help prevent excess weight gain and obesity as most vegetables and fruit are low in energy (kilojoules) and generally high in dietary fibre compared with many other foods
 - protects the body against non-communicable diseases such as heart disease, stroke and some cancers.

Evidence points particularly to the value of non-starchy vegetables in the diet. For example, the World Health Organization's recommendation on vegetables does not include starchy vegetables (WHO 2003). However, potatoes and other starchy vegetables, such as kūmara and taro, are traditional staples in many people's diets. Some also have cultural significance. Starchy vegetables provide nutrients such as carbohydrate and some vitamins and minerals to the diet and can be filling. However, they tend to be denser in energy (kilojoules) so it is healthier to have more non-starchy vegetables than starchy vegetables on the plate.

Eating Statement 1

Enjoy a variety of nutritious foods every day including: grain foods, mostly whole grain and those naturally high in fibre



Reasons for the recommendation

Eating whole grain and high-fibre grain foods is linked with a lower risk of cardiovascular disease, type 2 diabetes, weight gain and some cancers, such as bowel cancer.

- Whole grain and high-fibre grain foods provide energy (mainly from carbohydrate but also some protein), dietary fibre, vitamins including B group vitamins (except B12) and vitamin E (found particularly in wheatgerm) and minerals, such as magnesium, calcium, iron, zinc and selenium.
- Grains are a big part of diets around the world and for this reason they are considered dietary staples. They are generally affordable and easily available foods that give people some of the nutrients they need.
- Whole grains and those high in naturally occurring fibres provide better health benefits than more refined grains. In the definition of the Food and Agriculture Organization of the United Nations and World Health Organization (Mann et al 2007), dietary fibre is described as, and limited to, polysaccharides that are part of the plant cell wall, indicating fibre that is a naturally occurring part of the plant.

Background

Grain foods, also known as cereals, come from plants. Some of the grains that New Zealanders commonly eat are wheat, rice, oats, rye and barley. Less commonly eaten are millet, maize (corn) and more recently spelt (a type of wheat).

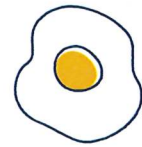
Defining whole grains

There is no one universally agreed definition of the term whole grain. Many people use 'whole grains' or 'intact grains' to mean grains that still have their key parts intact – that is, the bran, endosperm and germ (see [Figure 2](#)). In the Food Standards Code⁸ definition (FSANZ 2014), whole grain includes milled, dehulled, cracked, flaked or ground grains that contain the bran, endosperm and germ in the same proportions as the intact grain.

⁸ The Food Standards Code sets standards for foods commercially available in New Zealand and Australia, which must be followed by law. The Code contains standards for food additives, food safety and labelling.

Eating Statement 1

Enjoy a variety of nutritious foods every day including: some legumes¹³, nuts, seeds, fish and other seafood, eggs, poultry and/or red meat with the fat removed



Reasons for the recommendation

Eating patterns that include legumes, nuts, fish and other seafood are linked with a lower risk of cardiovascular disease, type 2 diabetes, weight gain and some cancers.

- Legumes, nuts and seeds are rich in nutrients and high in fibre and are a source of protein. Some types of nuts are useful sources of specific nutrients (eg, almonds provide calcium and Brazil nuts provide selenium). Nuts are high in unsaturated fats, but eating a small amount (around 30 g) each day should not cause excess weight gain, especially if you eat them instead of other, less healthy foods (Tey et al 2012).
- Fish and seafood are good sources of iodine.
- Oily fish such as salmon, tuna, mackerel and sardines and some seafood like mussels are good sources of omega-3 fatty acids. Omega-3 is linked with a lower risk of heart disease and stroke (Heart Foundation 2012).
- Eggs provide useful nutrients and can be part of a healthy diet for adults in general (Heart Foundation 2016).
- Poultry (eg, chicken) is a good source of protein and some minerals, including iron and zinc. Poultry has a variable fat content depending on the type of bird, but as most of it occurs in and around the skin, it is easy to remove (Mann and Truswell 2012).
- Red meat is an excellent source of key nutrients like iron (in an easily absorbed form) as well as zinc. Low iron levels are a problem for some New Zealanders, particularly young women. (Also see advice for pregnant and breastfeeding women under 'Folic acid, iodine, vitamin D and iron, page 38.)
- The World Cancer Research Fund (WCRF) reports that eating more than 500 g of cooked red meat (equivalent to 700–750 g when raw) each week is linked with a higher risk of colorectal cancer (WCRF and AICR 2007, 2011).
- Eating processed meat (eg, salami, bacon, ham and luncheon) is also linked with a higher risk of colorectal cancer (WCRF and AICR 2007, 2011). In addition, processed meats can be high in fat and salt.

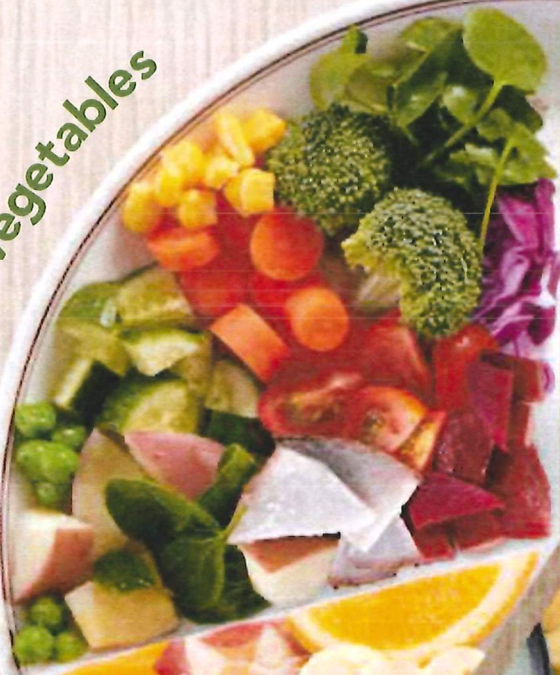
¹³ Legumes include lentils, split peas, chickpeas and cooked dried beans (eg, kidney beans, baked beans).



Water

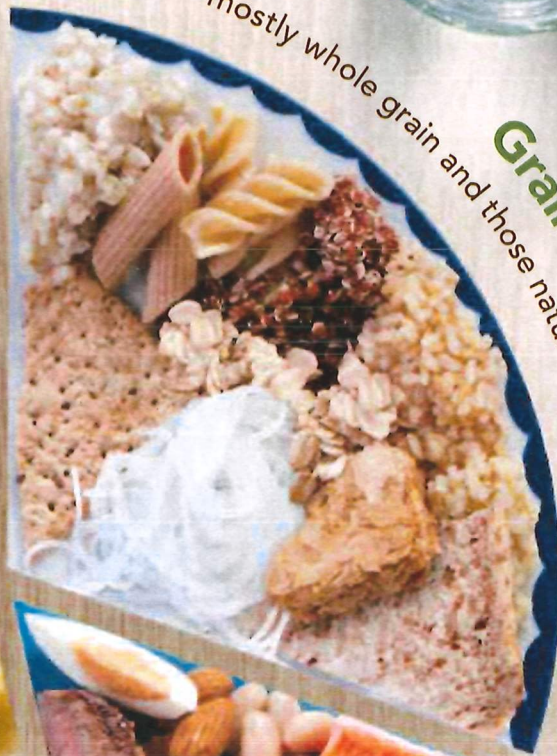
make it your
drink of choice

Vegetables



Grains

mostly whole grain and those naturally high in fibre



Fruit



Milk & milk products

mostly low and reduced fat



Legumes (e.g. lentils and beans),
seeds, fish, seafood, eggs,
poultry and/or lean red meat





Kegel Exercises for Men

What Will I Learn By Reading This Booklet?

When you have prostate cancer surgery or radiation therapy the muscles that help you control your urine flow may be weakened. When this happens you may have **incontinence**.

Incontinence is when you leak or pass urine when you do not want to. This is a very common **side effect** or unwanted change of prostate cancer treatment. The good news is that there is a simple exercise, called a **Kegel (Key-gul) exercise**, you can do to help strengthen your muscles. This exercise will help you have more control over your urine flow after your prostate cancer treatment. In this booklet you will learn:

- What a Kegel exercise is
- Why you should do Kegel exercises
- How to find your pelvic floor muscles
- How often you should do your Kegel exercises

It is important for you to think about and plan how you will take care of yourself before and after your prostate cancer treatment so that you can keep doing as many of your normal activities as possible.

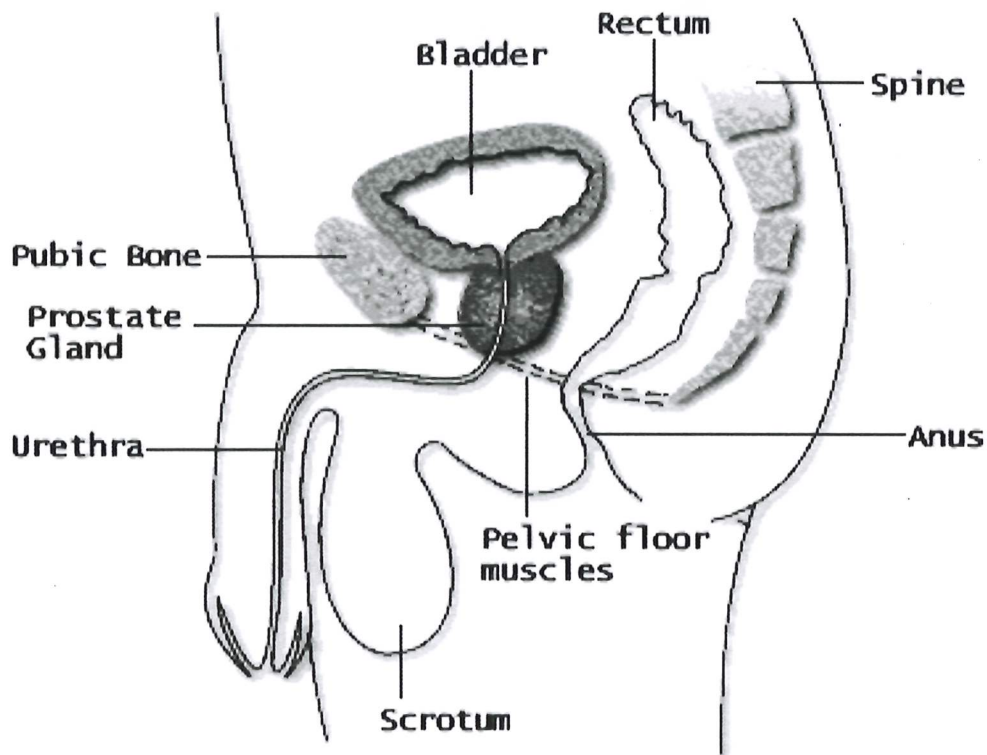
What Are Pelvic Floor Muscles?

Your **pelvic floor muscles** are a network of muscles that support your bladder and help you control your urine flow.

There are three pelvic muscles:

1. The **bladder**. The bladder is a muscle shaped like a balloon and holds your urine.
2. The **sphincter muscles**. These muscles help you open and close your urethra, the tube that drains urine from your bladder. And,
3. The **pelvic floor muscle** or **PC muscle** supports your bladder and **rectum** and helps control your urine flow.

Picture of Pelvic Floor Muscles



What Are Kegel Exercises?

Kegel exercises are easy exercises you can do before and after your prostate cancer treatment to help strengthen your pelvic floor muscles. These muscles help control your urine flow. Kegel exercises are one of the most effective ways of controlling incontinence without medication or surgery.

Why Should I Do Kegel Exercises?

The **prostate** is a gland, about the size of a walnut, located under the **bladder** surrounding the upper part of the urethra. The **urethra** is a tube that carries urine through the penis to the outside of the body. There are many muscles that surround the prostate gland. These muscles may be weakened during your prostate cancer treatment. This may cause you to have urine leakage also known as incontinence. Building up the strength in your pelvic floor muscles can help you gain better control of your bladder and urine flow. Remember, that just as it takes time to build your biceps and strengthen any other muscle in your body, it takes time to strengthen muscles in your pelvic floor.

How Do I Find My Pelvic Muscles?

In order to help strengthen your pelvic floor muscles, it is important that you take the time to make sure you are exercising the right muscles. It may take you several tries to find your pelvic muscles. So, take your time.

There are several ways that you can find your pelvic floor muscles.

One way is to: Try to stop and start your urine stream while you stand at your toilet to urinate (pee). Try to do this two or three times.

Another way to find your pelvic floor muscles is to:

1. Imagine that someone walks in to your bathroom while you are urinating (peeing) and you need to stop your urine flow.
2. Try to stop your urine flow.

The muscles you use to stop your urine flow are your pelvic floor muscles. These are the muscles you want to strengthen before and after your prostate cancer treatment.

How Do I Do A Kegel Exercise?

Now that you have located your pelvic floor muscles, you can exercise them even when you do not have to urinate (pee) by following these simple steps:

1. Tighten and hold your pelvic floor muscles for five seconds
2. Relax your pelvic muscles

You have just done one Kegel exercise. You should plan to do 10 to 20 Kegel exercises three to four times each day.

Another way to tighten your pelvic floor muscles is to:

1. Squeeze the muscles in your **anus** (like you are holding a **bowel movement**).
2. Relax your pelvic floor muscles after each attempt.
3. Repeat this exercise 10 to 20 times.

When you do your Kegel exercises, remember:

- **Do not** hold your breath
- **Do not** push down. Squeeze your muscles together tightly and imagine that you are trying to lift this muscle up.
- **Do not** tighten the muscles in your stomach, buttocks, or thighs.
- **Relax** your pelvic floor muscles between each squeeze.

How Often Should I Do Kegel Exercises?

When you first start doing your Kegel exercises, you may not be able to repeat the exercise 10 to 20 times. This is ok. It is much better for you to do fewer Kegel exercises that make your pelvic floor muscles stronger, than to do more exercises that do not work the muscle in the right way. As you get better at doing your Kegel exercises, slowly increase the number of times you repeat the exercise until you reach 20. Your goal should be to do 20 Kegel exercises three to four times each day.

The great thing about Kegel exercises is that you can do them anytime. No one can tell that you are doing these exercises. You can do your Kegel exercises before you get up in the morning, at lunchtime, at suppertime, and at bedtime. You can do them while you are watching TV or reading. Kegel exercises can help both men and women. Some men's partners do the exercises with them. The more you do them, the stronger your pelvic floor muscles will become.

When Should I Do My Kegel Exercise?

It can take six weeks or longer to strengthen your pelvic floor muscles so it is best to start doing your Kegel exercises before you have prostate cancer treatment. This will help you become better at doing the Kegel exercises and strengthen your pelvic floor muscles before your treatment starts. Remember, your pelvic floor muscles are like any other muscle in your body. It takes regular exercise and time to strengthen them.

Can I Do Kegel Exercises If I Have A Catheter?

A **catheter** is a thin rubber tube placed in your body to drain urine from your bladder out through your penis. Do not do any Kegel exercises if you have a catheter in your penis. Start doing your Kegel exercises according to the instructions above as soon as the catheter is taken out of your penis. After your catheter is removed you may experience some urine leakage (incontinence) when you stand up, cough, sneeze, laugh, or lift something. You will probably need to use incontinence pads for a while. However, doing Kegel exercises may help you control your urine flow sooner. If you have urine leakage when you stand up, cough, sneeze, laugh, or lift something, try doing a Kegel exercise. This may keep you from leaking urine.

Will My Urine Leakage Stop If I Do Kegel Exercises?

Most men gain control of their urine leakage (incontinence) within nine to 12 months after their surgery. Studies with men show that Kegel exercises help lessen urine leakage. However, each man's healing time is different.

Do not become discouraged if you have urine leakage. If you do your Kegel exercises on a daily basis, you can expect to see some results. You may have a great improvement or you may help keep your urine leakage from getting worse. You will need to continue doing your Kegel exercises each day so that your pelvic floor muscles stay strong.

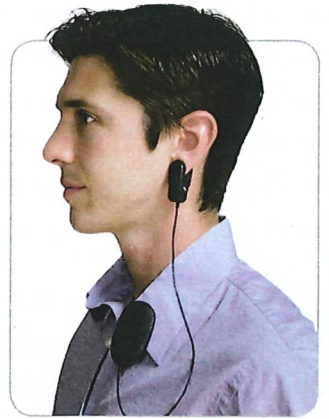
If you have any questions or concerns about urine leakage (incontinence), how to do a Kegel exercise, or if you would like to know about other things that may help with urine leakage, please talk to your doctor or your healthcare team.

Inner Balance™ Bluetooth® Instructional Sheet

WHAT IS IT? WHAT DOES IT DO?

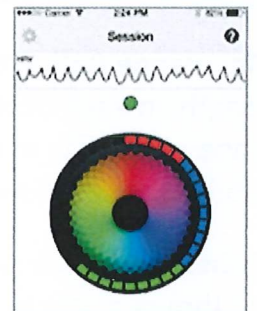
It can be easy to feel overwhelmed, anxious and scattered. The Inner Balance™ technology trains you to shift and replace emotional stress with greater balance and coherence.

It analyzes and displays your heart rhythms, measured by Heart Rate Variability (HRV), which indicates how emotional states are affecting your nervous system. (For more science, see page 2). The technology instructs users such as yourself to gain greater resilience through the practice of simple self-regulation techniques during the day.



HOW TO DO IT?

- ▶ Clip the black ear sensor to your earlobe and clip the black pod to your clothing. Turn on the button on the black pod.
- ▶ Press the arrow button at the bottom of the screen to start an Inner Balance session. In the beginning, it will calibrate for 15-20 seconds to make sure you have a good pulse reading (at the top of the screen).
- ▶ Next, focus on the area of your heart or chest area and follow the breath pacer (either a colored mandala or as you swipe through other screens, you will see other breathing pacers). Inhale for 3-5 seconds and exhale for 3-5 seconds. HRV patterns will show up at the top of the screen.
- ▶ As you sustain your deep breathing, try to engage a positive emotion or recall a time when you felt care, calm or appreciation. Try to practice this simple technique (called Quick Coherence®) for 1-3 minutes. The more coherent (in sync) you are, a blue or green color will show up indicating a medium or high coherence level.



ENDING A SESSION

To end a session, single tap the screen and then press on the small box at the bottom.

YOUR RESULTS

Once completed, the session review screen will display your results. When your heart rhythms (HRV patterns) are more ordered, you receive a higher score in the **medium** or **high** coherence scores. "Achievement" reflects how you were doing over the whole session. The more time in medium or high Coherence, the higher the achievement score. (For more explanation on results, see page 2).



HeartMath® Institute
expanding heart connections

THE SCIENCE BEHIND THE INNER BALANCE TECHNOLOGY

The Inner Balance technology measures Heart Rate Variability (HRV). Heart rate variability is a measure of the beat-to-beat changes in heart rate. Scientists and physicians consider HRV to be an important indicator of health and fitness. As a marker of physiological resilience and behavioral flexibility, it reflects our ability to adapt effectively to stress and environmental demands. The normal variability in heart rate is due to the synergistic action of the two branches of the autonomic nervous system (ANS)—the part of the nervous system that regulates most of the body's internal functions. The sympathetic nerves act to accelerate heart rate, while the parasympathetic (vagus) nerves slow it down.

Many factors affect the activity of the ANS, and therefore influence HRV. These include our breathing patterns, physical exercise, and even our thoughts. Research at the HeartMath Institute has shown that one of the most powerful factors that affects our heart's changing rhythm is our feelings and emotions (*American Journal of Cardiology*, 1995).

Heart rhythm patterns during different emotional states. These graphs show examples of real-time heart rate variability patterns (heart rhythms) recorded from the same individual experiencing different emotions. The incoherent heart rhythm pattern shown in the top graph, characterized by its irregular, jagged waveform, is typical of stress and negative emotions such as anger, frustration, and anxiety. The bottom graph shows an example of the coherent heart rhythm pattern that is typically observed when an individual is experiencing a sustained positive emotion, such as appreciation, compassion, or calm. The coherent pattern is characterized by its regular, sine-wave-like waveform.

INTERPRETING YOUR RESULTS

The technology includes an infrared sensor connected to your iPhone, iPad or Android device. Session screens plot the pulse, HRV, power spectrum and coherence over time.

Coherence Ratios are a session summary of the amount of time, as a percentage of the total session length, in each of the three Challenge Level ranges. If your session was 3 minutes long and your coherence score was in the low, medium, and high range for 1 minute each then the coherence ratio display would show 33 low, 33 medium, and 33 high.

Coherence score is a measure of the degree of coherence in the heart rhythm pattern. A coherent heart rhythm is a stable regular repeating rhythm resembling a sine wave at a single frequency between 0.04-0.24 Hz (3-15 cycles per minute). The scoring algorithm continuously monitors the most current 64 seconds of heart rhythm data and updates the score every 5 seconds. The more stable and regular the heart rhythm frequency, the higher the coherence score. Scores range from 0-16. With practice and regular use, you will begin to notice your own normal coherence score level and how it fluctuates when you have more or less focus and when you experience greater levels of inner balance.

Achievement Score is the total of all coherence scores awarded every 5 seconds during a session. The scoring algorithm updates your coherence score every 5 seconds during an active session and adds them together throughout the session giving you a sum of coherence scores for the entire session. You may want to set a daily achievement goal for yourself. An achievement score of 300 points per day is a good place to start. This could be accumulated in a single session or split across two or more sessions depending on your schedule or preferences.

Challenge Level settings are thresholds or levels of coherence. With an 80% or more in high coherence, you are encouraged to move up a challenge level (click on settings in upper left hand corner).

The Impact of Stress and the Basics of Heart Coherence



What is stress?

Stress is a term used to describe the wear and tear the body experiences in reaction to everyday tensions and pressures. Change, illness, injury or career and lifestyle changes are common causes of stress. It's how we respond, however, to the emotional pressure and tension we feel from the little everyday hassles—rush-hour traffic, waiting in line and too many e-mails—that do the most damage.

How does stress affect health?

Stress affects people physically, mentally and emotionally. According to the American Institute of Stress, up to 90% of all health problems are related to stress. Too much stress can contribute to and agitate many health problems, including heart disease, high blood pressure, stroke, depression and sleep disorders.

How stressed are you?

Everyone responds to stress differently. Take this quick test to see if you are experiencing stress. The items you check will reveal how much stress is affecting your life.

- ☐ I feel overly tired or fatigued.
- ☐ I often am nervous, anxious or depressed.
- ☐ I have sleep problems.
- ☐ I have repeated headaches or minor aches and pains.
- ☐ I worry about my job, finances, relationships.

What can I do about it?

The first step is to understand how stress works. It's not the events or situations that do the harm, but rather how you respond to those events. More precisely, it's how you feel about them that determines whether you are stressed.

Emotions, or feelings have a powerful impact on the human body. Positive emotions like appreciation, care and love not only feel good, they are good for you. They help your body's systems synchronize and work better, as a well-tuned car. Research at the Institute of HeartMath has shown that when you intentionally shift to a positive emotion, heart rhythms immediately change. This shift in heart rhythms creates a favorable cascade of neural, hormonal and biochemical events that benefit the entire body. The effects are both immediate and long lasting.

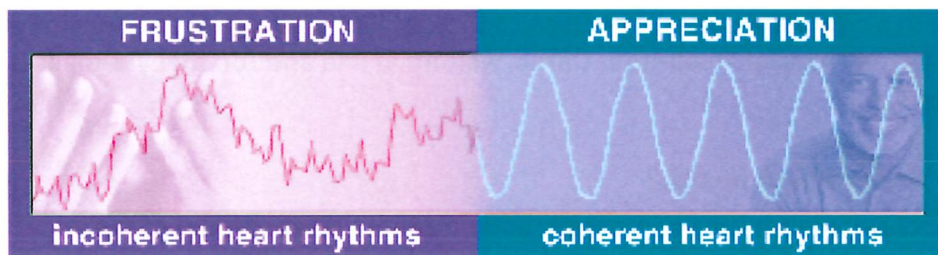
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The Impact of Stress and the Basics of Heart Coherence



How does it work?

When you're stressed, your body is out of sync. Negative/depleting emotions we feel when stressed such as anger, frustration, anxiety and worry lead to increased disorder in heart rhythms and the nervous system. In contrast, positive/renewing emotions like joy, appreciation, care and kindness create harmony in heart rhythms and the nervous system. Other bodily systems sync up to this rhythm, a process scientists call coherence. Because coherence leads to more mental clarity, creativity and better problem-solving abilities, it's easier to find solutions and better ways of handling stressful situations.



How can I create and practice physiological coherence?

The HeartMath® System introduces you to a powerful tool to help you increase physiological coherence. Use the Quick Coherence® Technique in your daily life to reduce the effects of stress and the emWave® technology to enhance your coherence practice.

The Quick Coherence® Technique

The Quick Coherence Technique is an easy way to interrupt the stress response and quickly bring your system into coherence. Practice this tool four or five times a day, every day for a minimum of three weeks. Use it as a quick stress manager prior to or during challenging events or situations. Good times to practice the Quick Coherence steps are first thing in the morning, before going to sleep at night and break time in the middle of the day, but you can use it anytime you want to rebalance or get an energy boost.

The Steps

Step 1. Focus your attention in the area of the heart. Imagine your breath is flowing in and out of your heart or chest area, breathing a little slower and deeper than usual. Find an easy rhythm that's comfortable.

Step 2. As you continue heart-focused breathing make a sincere attempt to experience a regenerative feeling such as appreciation or care for someone or something in your life.

Suggestion: Try to re-experience the feeling you have for someone you love, a pet, a special place, an accomplishment, etc., or focus on a feeling of calm and ease.

Quick Steps:

1. Heart-Focused breathing
2. Activate a positive or renewing feeling

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Quick Coherence® Technique Basics



The Heart-Focused Breathing™ step shifts your attention away from your depleting thoughts to your heart or chest area increases coherence in your system. Because changing the rhythm of your breathing changes the rhythm in your nervous system, when you breathe deeper and slower, 5 seconds in and 5 seconds out or 6 complete breaths per minute, you bring more order to your nervous system and the rhythm of your heart. Doing this anytime you feel stress has a powerful soothing effect on the entire body and will help you start to feel calmer.

Activate a positive or renewing emotion. Making a sincere attempt to experience a renewing feeling such as appreciation or care for someone or something in your life helps you sustain the coherence started with the Heart-Focused Breathing step, without having to remain focused on your breath. Many people find that when they experience positive feelings like care, love or appreciation while practicing heart-focused breathing through the heart or chest area, they immediately feel uplifted and regenerated. We call these *heart feelings* because they change how your heart is beating and because people often describe these feelings or emotions as *coming from the heart*.

If you are accustomed to having more negative, depleting feelings than positive, renewing feelings, breathing this way may feel uncomfortable at first because it's unfamiliar. For now, simply make a sincere effort to feel and hold the renewing emotions by recalling a positive or fun time in your life and reexperiencing it.

If it's hard for you to recall anything positive, write down some occasions when you felt an uplifting, positive feeling. This will help you remember how to recall one of them when you practice the Quick Coherence Technique. Even a sincere attempt to feel a positive emotion will reduce the drain and replenish your system.

Practice makes it easier to sustain coherence for longer periods of time. Coherence will become more natural and familiar to your system, making it much easier to remain calm and balanced in challenging or stressful times.

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Stress Facts

Understanding the mechanics of stress gives you the advantage of being more aware of and sensitive to your own level of stress and knowing when and how to take proactive steps. This increased awareness also helps you to better care for your family, friends and colleagues.

Here are a few stress facts that many people are unaware of:

■ **Fact #1: Your body doesn't care if it's a big stress or a little one.**

The human body doesn't discriminate between a BIG stress, or a little one. Regardless of the significance, stress affects the body in predictable ways. A typical stress reaction, which most of us experience dozens of times each day, begins with a cascade of 1,400 biochemical events in your body. If these reactions are left unchecked we age prematurely, our cognitive function is impaired, our energy is drained, and we are robbed of our effectiveness and clarity.

■ **Fact #2: Stress can make smart people do stupid things.**

Stress causes what brain researchers call "cortical inhibition." The phenomenon of cortical inhibition helps to explain why smart people do dumb things. Simply said, stress inhibits a small part of your brain and you can't function at your best. When we are in coherence – a state where we are cognitively sharp, emotionally calm, and we feel and think with enhanced clarity – the brain, heart and nervous system are working in harmony. This state of coherence facilitates our cognitive functioning – we are actually operating at peak performance mentally, emotionally and physically.

■ **Fact #3: People can become numb to their stress.**

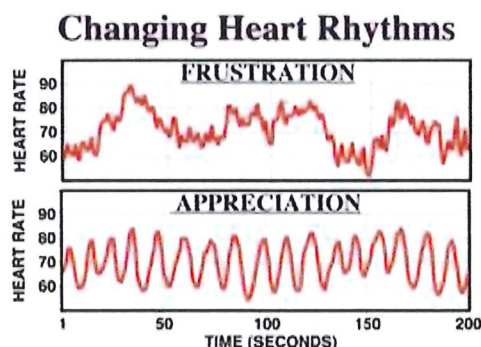
We can be physiologically experiencing stress yet mentally numb to it because we've become so accustomed to it. Some have become so adapted to the daily pressures, irritations and annoyances of life that it starts to seem normal. Yet the small stresses accumulate quickly and we may not realize how much they're impairing our mental and emotional clarity and our overall health until it shows up as a bad decision, an overreaction or an unwanted diagnosis at the doctor's office.

■ **Fact #4: We can control how we respond to stress.**

We don't need to be victims to our own emotions, thoughts and attitudes. We can control how we respond to stress and we can become more sensitive to stressful situations and how they are affecting us before it manifests as a physical, mental or emotional complaint. There are simple, scientifically validated solutions to stress that empower people to rewire their own stress response.

■ **Fact #5: The best strategy is to handle stress in the moment.**

The best way to manage stress is to deal with it the very moment you feel it come up. Millions of Americans unsuccessfully use the binge-and-purge approach when it comes to stress. They stress out all day, believing that they can wait until later to recover when they go to an evening yoga class, go to the gym or chill out when they take the weekend off. Unfortunately, when we put off going for our own inner balance our bodies have already activated the stress response and it's our health that suffers.



HeartMath's research shows how emotions change our heart rhythm patterns. Positive emotions create coherent heart rhythms, which look like rolling hills – it's a smooth and ordered pattern. In contrast, negative emotions create chaotic, erratic patterns. Using a heart rhythm monitor, you can actually see your heart rhythms change in real time as you shift from stressful emotions like anger or anxiety to positive feelings like care or appreciation. Coherent heart rhythm patterns facilitate higher brain function, whereas negative emotions inhibit a person's ability to think clearly. Coherent heart rhythms also create a feeling of solidity and security.

Dates:

	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Heart Rate Monitor (minutes)							
Morning Pelvic Floor (y/n)	☐	☐	☐	☐	☐	☐	☐
Noon Pelvic Floor (y/n)	☐	☐	☐	☐	☐	☐	☐
Evening Pelvic Floor (y/n)	☐	☐	☐	☐	☐	☐	☐
Exercise (minutes)		Strength Workout A			Strength Workout B		
Connection/ Intimacy (type)							

Weight: _____

Dates:

	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Heart Rate Monitor (minutes)							
Morning Pelvic Floor (y/n)	☐	☐	☐	☐	☐	☐	☐
Noon Pelvic Floor (y/n)	☐	☐	☐	☐	☐	☐	☐
Evening Pelvic Floor (y/n)	☐	☐	☐	☐	☐	☐	☐
Exercise (minutes)		Strength Workout A			Strength Workout B		
Connection/ Intimacy (type)							

Weight: _____